



Software Requirements Specification

for

Aion Pharmacy

Version 1.0 approved

Prepared by Group D

Aion Pharmacy

2026-05-14

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Revision History

Name	Date	Reason For Changes	Version
Group D	2026-05-14	Initial SRS based on implemented system	1.0

Software Requirements Specification (SRS)

Aion Pharmacy

- Version: 1.0
 - Creation date: 2026-05-14
 - Revision date: 2026-05-14
 - Product: Retail pharmacy management system
 - Project: Aion Pharmacy (aion-laravel)
 - Prepared by: Group D
 - Approver: TBD
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1. Overview

1.1 Purpose

This SRS defines the implemented software requirements for the Aion Pharmacy retail pharmacy management system. It is intended for project stakeholders, developers, testers, and maintainers who need a clear, complete view of system scope and behavior.

This document also provides a short narrative overview and structured requirements so readers can understand how the system supports daily pharmacy work, then verify each feature against the detailed lists.

1.2 Business Context

Aion Pharmacy supports day to day pharmacy operations for small to medium retail pharmacies. It centralizes customer records, medication catalog management, inventory visibility, prescriptions, point of sale checkout, and basic reporting for operational oversight.

At a high level, the system helps staff record patient details, dispense medications, and complete sales while keeping inventory and audit trails consistent for reporting and accountability.

1.3 Scope

The system provides role based pharmacy operations for staff (admins and pharmacists). It includes authentication, customer management, medication catalog management, inventory visibility with admin only adjustments, prescription creation and status updates, sales checkout with stock deduction, stock movement audit history, reporting, and staff user management.

Out of scope for v1.0 (explicitly deferred): - Supplier procurement or purchase orders - Multi branch inventory allocation

Not applicable to this system: - Patient self service portals - Insurance billing and claims

1.4 User Characteristics

- Admin
- Function: Operations oversight, inventory adjustments, reporting, staff management
- Location: In pharmacy back office
- Device: Desktop or laptop browser
- Technical proficiency: Moderate, familiar with business applications
- Usage frequency: Daily, with periodic reporting
- Typical concurrent count: 1 to 2
- Pharmacist
- Function: Customer intake, prescriptions, sales checkout, inventory visibility
- Location: Front desk
- Device: Desktop or laptop browser
- Technical proficiency: Basic to moderate
- Usage frequency: Continuous during shifts
- Typical concurrent count: 1 to 5
- Customer (indirect)
- Function: Receives service from staff; does not access the system directly
- Accessibility considerations: Staff facing UI should support clear labels and keyboard navigation

1.5 Group Members

This project was developed by the following group members.

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2. Assumptions, Dependencies, Constraints

2.1 Assumptions

- The pharmacy has stable local network access to the application host.
- Staff are trained on basic web application usage.
- Inventory quantities are managed within the system, not in parallel systems.

2.2 Dependencies

- Web runtime: Laravel application, PHP runtime, Blade UI
- Database: MySQL compatible relational database
- Optional: Docker runtime for containerized deployment
- Client: Modern desktop browsers (Chrome, Edge, Firefox)

2.3 Constraints

- Role based access is enforced by server side authorization gates.
 - Pharmacists cannot perform manual inventory adjustments or manage users.
 - Admin only access to stock movement audit logs and reports.
 - All core operations require authentication.
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3. Requirements

3.1 Business Requirements

The business requirements summarize the outcomes the system must achieve for pharmacy operations, focusing on safe dispensing, accurate inventory, and clear accountability. - BR-1: The system shall support daily pharmacy operations for customer care, dispensing, and sales checkout. - BR-2: The system shall maintain an auditable record of stock changes and sales transactions. - BR-3: The system shall separate permissions for admins and pharmacists to reduce operational risk. - BR-4: The system shall provide basic reports for sales and stock oversight.

3.2 Functional Requirements

3.2.1 Authentication and Profile

- FR-1: The system shall allow users to register, log in, log out, and reset passwords.
- FR-1a: Registration shall require name, email, and password confirmation.
- FR-1b: Self registered accounts shall be assigned the pharmacist role.
- FR-2: The system shall allow authenticated users to update their profile details (name, email).
- FR-3: The system shall allow authenticated users to delete their own account with password confirmation.

3.2.2 Dashboard

- FR-4: The system shall provide a role aware dashboard with summary counts and recent activity.
- FR-4a: The dashboard shall display counts for customers, medications, and today's prescriptions.
- FR-4b: Users with sales access shall see low stock alerts and today's sales totals.
- FR-4c: Admins shall see recent stock movements; all staff shall see recent sales and prescriptions based on access.
- FR-5: The system shall show low stock indicators when quantity on hand is less than or equal to reorder level.

3.2.3 Customer Management

- FR-6: The system shall allow staff to create, edit, list, and delete customer records.
- FR-6a: The customer list shall be paginated with 12 records per page.
- FR-6b: The customer list shall support search by first name, last name, phone, or email.
- FR-6c: Customer creation shall require first name, last name, date of birth (before today), and sex.
- FR-6d: Optional customer fields shall include phone, email, address, medical history, allergies, and conditions.
- FR-7: The system shall store customer demographics, contact details, and optional medical notes.

3.2.4 Medication Catalog

- FR-8: The system shall allow staff to create, edit, list, view, and delete medications.
- FR-8a: The medication list shall be paginated with 12 records per page.
- FR-8b: The medication list shall support search by name or SKU and filter by status.
- FR-8c: The medication list shall support table and card view modes.
- FR-9: The system shall enforce unique medication SKU identifiers.
- FR-10: Medication creation shall require SKU (max 64), name (max 255), unit type (max 32), unit price (≥ 0), and status (active or inactive).
- FR-10a: Optional medication fields shall include dosage form (max 64), strength (max 64), reorder level (≥ 0), and image (≤ 2 MB).
- FR-10b: When a medication image is replaced or deleted, the prior file shall be removed from storage.

3.2.5 Inventory Visibility

- FR-11: The system shall display current quantity on hand for each medication.
- FR-11a: Inventory list shall support search by medication name or SKU.
- FR-11b: Inventory list shall support filtering to show only low stock items.
- FR-11c: Inventory list shall support table and card view modes.
- FR-12: The inventory detail view shall show quantity on hand, reserved quantity, and recent stock movements (latest 12).

3.2.6 Inventory Adjustments (Admin Only)

- FR-13: The system shall allow admins to record stock adjustments with quantity, movement type, and optional notes.
- FR-13a: Movement types shall include in, out, and adjustment.
- FR-13b: Adjustment direction shall be increase or decrease when movement type is adjustment.
- FR-13c: Quantity shall be an integer of 1 or greater.
- FR-13d: Decreases shall be blocked when the quantity exceeds stock on hand.
- FR-14: The system shall log each adjustment as a stock movement entry with reference type manual.

3.2.7 Prescriptions

- FR-15: The system shall allow staff to create prescriptions for customers with one or more items.
- FR-15a: Prescription creation shall require a customer, and at least one medication item.
- FR-15b: Each item shall require medication id and quantity (≥ 1), with optional dosage instructions.
- FR-15c: A prescription number shall be generated in the format RX-YYYYMMDD-XXXXXX.
- FR-16: The system shall allow staff to view prescription details and update prescription status.
- FR-16a: Prescription list shall be paginated with 20 records per page and filterable by status.
- FR-17: The system shall store prescription items with dosage instructions and price snapshots.

3.2.8 Sales Checkout

- FR-18: The system shall allow staff to create sales with one or more line items.
- FR-18a: Sale creation shall require at least one medication item.
- FR-18b: Each item shall require medication id and quantity (≥ 1).
- FR-19: The system shall support optional customer association with sales.
- FR-20: The system shall support payment method selection and discount and tax inputs.
- FR-20a: Payment methods shall be cash, card, or mobile.
- FR-20b: Discount and tax shall be numeric values ≥ 0 and applied at the sale level.
- FR-21: The system shall deduct inventory quantities for items sold and log stock movements.
- FR-21a: Sale creation shall fail with a validation error if any item quantity exceeds stock on hand.
- FR-22: The system shall provide a receipt style sale detail view.
- FR-22a: Sale list shall be paginated with 20 records per page and filterable by status.

3.2.9 Stock Movement Audit (Admin Only)

- FR-23: The system shall list historical stock movements with filtering and pagination.
- FR-23a: Stock movements list shall be paginated with 20 records per page.
- FR-23b: Stock movements list shall be filterable by movement type.

3.2.10 Reports (Admin Only)

- FR-24: The system shall provide sales summary reports.
- FR-24a: Reports shall include sales trends over a six month analytics window.
- FR-25: The system shall provide stock summary reports.
- FR-25a: Reports shall include low stock counts and movement summaries.
- FR-26: The system shall provide report export endpoints for overview, patients, sales, and stock.
- FR-26a: Export format shall be CSV with UTF-8 encoding.
- FR-26b: Overview export shall include summary metrics, sales trend, patient growth, top medications, recent sales, and recent patients.
- FR-26c: Patients export shall include patient name, phone, sex, prescription count, sales count, and joined date.
- FR-26d: Sales export shall include sale number, date, customer, cashier, payment method, status, subtotal, discount, tax, and total.
- FR-26e: Stock export shall include medication name, SKU, on hand, reserved, reorder level, and stock status.

3.2.11 Staff Users (Admin Only)

- FR-27: The system shall allow admins to create, edit, list, and delete staff users.
- FR-27a: Users list shall be paginated with 12 records per page and filterable by role.
- FR-27b: Users list shall support search by name or email.
- FR-27c: Staff roles shall be limited to admin and pharmacist.
- FR-27d: User creation shall require name, email, role, and password confirmation.
- FR-27e: Password change on edit shall be optional.
- FR-28: The system shall prevent admins from deleting their own account.

3.2.12 Business Rules

- BRULE-1: Inventory quantity on hand shall not be reduced below zero by sales or adjustments.
- BRULE-2: Low stock shall be defined as quantity on hand $\leq$ reorder level when reorder level is set.
- BRULE-3: Prescription status values shall be draft, confirmed, dispensed, or cancelled.
- BRULE-4: Prescriptions shall not adjust inventory quantities.
- BRULE-5: Sales shall be recorded with status paid and do not support void or refund workflows.
- BRULE-6: Payment methods shall be limited to cash, card, and mobile.

- BRULE-7: Discount and tax shall be applied at the sale header level, not per line item.
- BRULE-8: There is no super admin role; multiple admins may exist.
- BRULE-9: Deleting customers, medications, and users shall perform a hard delete (no soft delete).

3.3 Logical Data Requirements

The system shall persist the following core entities: - Users: staff accounts with roles (admin, pharmacist) - Customers: demographic and medical details - Medications: catalog data and pricing - Inventory: quantity on hand per medication - Stock Movements: audit history for stock changes - Prescriptions and Prescription Items - Sales and Sale Items

Key data rules and constraints: - Customer first name and last name are required (max 100 each), date of birth is required, and sex is one of male, female, or other. - Medication SKU is required, unique, and max 64; unit price is numeric and ≥ 0 ; status is active or inactive. - Inventory quantity on hand is integer ≥ 0 ; reserved quantity defaults to 0. - Prescription items and sale items require medication id and quantity ≥ 1 . - Sales and prescription line items store unit price snapshots at time of creation.

Relationships and cardinality: - One customer has many prescriptions and many sales. - One prescription has many prescription items. - One sale has many sale items. - One medication has one inventory record and many sale and prescription items. - One stock movement references one medication and one user.

Data retention and deletion: - Records persist until explicitly deleted by staff. - Deletions are hard deletes; no soft delete is implemented for customers, medications, or users. - Stock movement, sale, and prescription records are retained unless removed via delete operations or cascading deletes.

Backup and archival: - Backup scheduling and archival policies are external to the application and must be defined operationally.

3.4 User Requirements

- UR-1 (Admin): Manage staff accounts and assign roles.
- UR-2 (Admin): Adjust inventory and review movement history.
- UR-3 (Admin): View sales and stock reports.
- UR-4 (Pharmacist): Register customers and manage patient details.
- UR-5 (Pharmacist): Create prescriptions and complete sales checkout.
- UR-6 (All staff): View inventory availability and low stock indicators.

3.5 Information Management Requirements

- IM-1: The system shall store all sales and prescription records for later retrieval.
- IM-2: The system shall store stock movement records for audit and reconciliation.
- IM-3: The system shall store customer medical notes (history, allergies, conditions) for clinical reference.

- IM-4: The system shall store medication status to support active and inactive visibility.

3.6 Systems Requirements

3.6.1 Performance Requirements

- PR-1: The system shall support at least 10 concurrent authenticated sessions without data inconsistency or request failure under normal operating conditions.
- PR-2: List views and dashboards shall load within 3 seconds on a local network with up to 10,000 medications and 50,000 sales records.
- PR-3: Sale checkout and inventory deduction shall complete within 2 seconds under normal load.

3.6.2 Quality Requirements

- QR-1: The system shall enforce role based access control for sensitive features.
- QR-2: The system shall validate inputs on create and update operations.
- QR-3: The system shall maintain auditability for inventory changes.
- QR-4: The system shall use database transactions for sale checkout and inventory adjustments.
- QR-5: The system shall reject sale submissions when stock on hand is insufficient.
- QR-6: The system shall log application errors and exceptions to a persistent log.
- QR-7: Authorization checks shall be implemented via reusable gates or middleware.
- QR-8: Validation errors shall be displayed inline with descriptive messages.
- QR-9: Destructive actions shall require user confirmation before execution.

3.6.3 Security Requirements

- SR-1: Passwords shall be hashed using a one way algorithm (bcrypt via Laravel Hash).
- SR-2: Sessions shall expire after a configurable inactivity timeout; the default shall be 120 minutes.
- SR-3: Failed login attempts shall be rate limited.
- SR-4: Password reset tokens shall expire after 60 minutes and be single use.
- SR-5: All forms shall include CSRF token validation.
- SR-6: User supplied inputs shall be escaped on render to mitigate XSS.
- SR-7: Database access shall use ORM or parameterized queries to prevent SQL injection.
- SR-8: The application shall be deployable behind HTTPS; HTTP shall redirect to HTTPS in production.
- SR-9: Server side authorization gates shall be applied to all sensitive routes.
- SR-10: Pharmacists shall not access admin gated routes even via direct URL entry.

3.6.4 Error Handling and Logging Requirements

- EH-1: The system shall display user friendly error messages for 403 and 404 responses.

- EH-2: Unhandled server errors shall not expose stack traces in production.
- EH-3: Application exceptions shall be logged with timestamp and user context when available.
- EH-4: Transaction failures during checkout or inventory adjustment shall rollback and display a recoverable error.

3.6.5 Availability Requirements

- AR-1: The system shall be available during pharmacy operating hours except for planned maintenance.
- AR-2: Planned maintenance windows shall be communicated to staff in advance.

3.7 Interfaces

- UI: Browser based interface built with Blade templates.
- Database: MySQL compatible relational database.
- File storage: Public storage for optional medication image paths (if configured).
- Exports: Report export endpoints produce CSV files with UTF-8 encoding.

3.8 Other Requirements

- OR-1: The system shall be deployable using Docker or a local PHP runtime.
- OR-2: The system shall support standard Laravel authentication scaffolding flows.

4. Requirements Traceability Matrix

The following matrix maps key requirements to business goals, implementation, and testing.

Requirement ID	Business Requirement	User Requirement	Implementation Reference	Test Case ID	Status
FR-1 to FR-3	BR-1	UR-6	Auth routes and profile actions (routes/auth.php, profile routes)	TC-001	Implemented
FR-4 to FR-5	BR-1	UR-6	DashboardController, dashboard view	TC-002	Implemented
FR-6 to FR-7	BR-1	UR-4	CustomerController, customers resource routes	TC-003	Implemented
FR-8 to FR-10	BR-1	UR-6	MedicationController, medications resource routes	TC-004	Implemented
FR-11 to FR-12	BR-1	UR-6	InventoryController index and show routes	TC-005	Implemented

Requirement ID	Business Requirement	User Requirement	Implementation Reference	Test Case ID	Status
FR-13 to FR-14	BR-2	UR-2	InventoryController adjustment routes (admin gate)	TC-006	Implemented
FR-15 to FR-17	BR-1	UR-5	PrescriptionController create, store, show, status routes	TC-007	Implemented
FR-18 to FR-22	BR-1	UR-5	SaleController create, store, show routes	TC-008	Implemented
FR-23	BR-2	UR-2	StockMovementController index route	TC-009	Implemented
FR-24 to FR-26	BR-4	UR-3	ReportController index, sales, stock, patients, export routes	TC-010	Implemented
FR-27 to FR-28	BR-3	UR-1	UserController resource routes	TC-011	Implemented

5. Glossary

- Admin: Staff user with elevated permissions to manage inventory, reports, and users.
 - Pharmacist: Staff user who performs sales and prescription workflows.
 - SKU: Stock Keeping Unit identifier for a medication.
 - Inventory Adjustment: Admin initiated manual change to stock quantity with a stated reason.
 - Stock Movement: Audit record of inventory changes.
 - Reorder Level: Threshold quantity at or below which a medication is flagged as low stock.
 - Prescription: Clinical record with medication line items for a customer.
 - Status (Prescription): Lifecycle state of a prescription (draft, confirmed, dispensed, cancelled).
 - Sale: Point of sale transaction with line items and payment details.
 - Receipt: Sale detail view summarizing items, totals, and payment method.
 - Soft Delete vs. Hard Delete: Soft delete flags records as deleted; hard delete removes them. This system uses hard deletes.
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6. References

- GitHub repository: <https://github.com/Aion67/aion-laravel.git>
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7. Revision History

Version	Date	Name	Description
1.0	2026-05-14	Project team	Initial SRS based on implemented system

8. Appendices

Appendix A. Domain Model Summary

The domain model includes users, customers, medications, inventory, stock movements, prescriptions, prescription items, sales, and sale items.